A2 Warm Up 6.3

1. Express in simplest radical form.

$$3\sqrt{6}+\sqrt{6}$$

$$\boxed{4\sqrt{6}}$$

2. Express in simplest radical form.

$$\sqrt{3}+\sqrt{108}$$

$$\boxed{7\sqrt{3}}$$

3. Express the product $\left(7+\sqrt{2}\right)\left(7+\sqrt{2}\right)$ in simplest form.

$$\boxed{51+14\sqrt{2}}$$

4. Express the product $\left(\sqrt{10}-\sqrt{3}\right)\left(\sqrt{10}+\sqrt{3}\right)$ in simplest form.

$$\boxed{7}$$

5. Multiply and express in simplest radical form:

$$(-6-\sqrt{45})(5+3\sqrt{5})$$

$$\boxed{-75-33\sqrt{5}}$$

6. Multiply and express in simplest radical form:

$$(-5-3\sqrt{6})(1+2\sqrt{6})$$

$$\boxed{-41-13\sqrt{6}}$$

7. Express as a fraction in simplest form with a rational denominator:

$$\frac{5}{2-\sqrt{6}}$$

$$\boxed{\frac{-10-5\sqrt{6}}{2}}$$

8. Express as a fraction in simplest form with a rational denominator:

$$\frac{7}{2+\sqrt{2}}$$

$$\boxed{\frac{14-7\sqrt{2}}{2}}$$

9. Express as a single fraction in simplest radical form with a rational denominator.

$$\frac{\sqrt{5}-\sqrt{12}}{\sqrt{10}+\sqrt{6}}$$

$$\boxed{\frac{11\sqrt{2}-3\sqrt{30}}{4}}$$

10. Express as a single fraction in simplest radical form with a rational denominator.

$$\frac{4-\sqrt{2}}{6+\sqrt{2}}$$

$$\boxed{\frac{13-5\sqrt{2}}{17}}$$